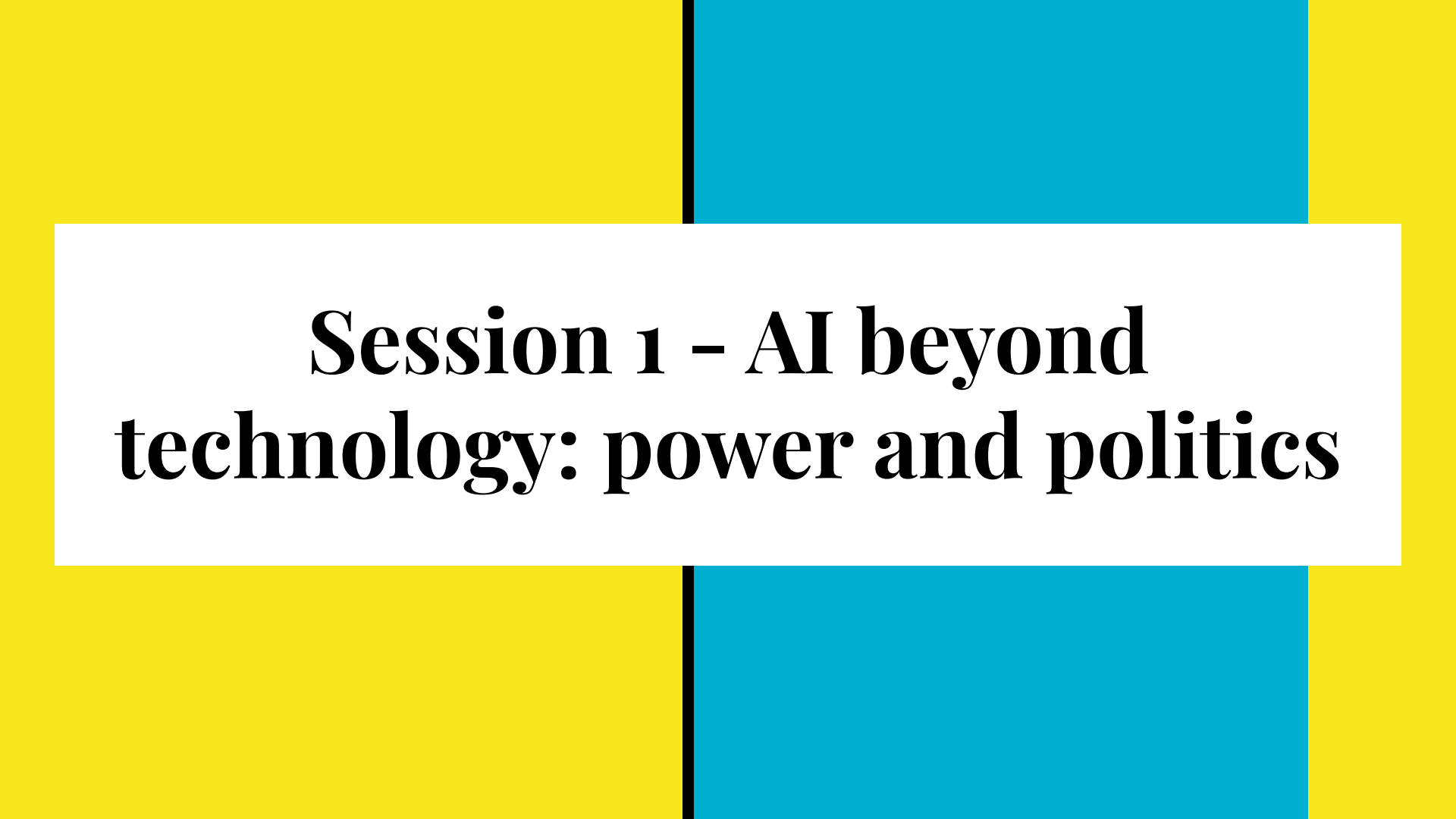


Impacts of Artificial Intelligence

KEDGE - Data Analytics for Business

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The background consists of a yellow field with two large blue squares, one in the top right and one in the bottom right. A black crosshair is centered on the page, with its vertical line passing through the blue squares and its horizontal line passing through the white text box.

Session 1 – AI beyond technology: power and politics

Reminder

Session 1 : AI beyond technology : power and politics

- CORBYN Z., 2021, « Microsoft's Kate Crawford: 'AI is neither artificial nor intelligent' », [The Observer | The Guardian](#), 6 juin 2021.
- HAO K., 2021, « Stop talking about AI ethics. It's time to talk about power. », [MIT Technology Review](#), 23 avril 2021.
- *Complementary reading* : CRAWFORD K., 2021, Atlas of AI: Power, Politics, and the Planetary Costs of Artificial Intelligence, New Haven, Yale University Press.



CRAWFORD K., 2021, Atlas of AI

Power, Politics, and the Planetary Costs
of Artificial Intelligence

Neither artificial nor intelligent

Earth

Labor

State

POWER

EXERCISE - it's your turn to work

(In groups): Read the two articles and then answer the following questions

- *(Use the guardian as a basis and then use the MIT article to complete)*

> What is the main objective of the book and its originality?

> Can you give an example cited by the author of the problematic uses of AI? Can you give an example of your own?

> What does Crawford mean when she says that we have to move from the question of ethics to the question of power? What do you think?

**Neither artificial nor
intelligent**

How to define AI

Various perceptions of AI

- average citizen versus expert deep learning

The chosen **definition sets a framework**

- how AI will be understood, measured, evaluated, regulated, etc.

Often presented by marketing as an **intelligent, artificial, technical** system, etc.

“Neither artificial nor intelligent”

we'll break it down



“In contrast, in this book I argue that AI is neither artificial nor intelligent. Rather, artificial intelligence is both embodied and material, made from natural resources, fuel, human labor, infrastructures, logistics, histories, and classifications. AI systems are not autonomous, rational, or able to discern anything without extensive, computationally intensive training with large datasets or predefined rules and rewards. In fact, artificial intelligence as we know it depends entirely on a much wider set of political and social structures. And due to the capital required to build AI at scale and the ways of seeing that it optimizes AI systems are ultimately designed to serve existing dominant interests. In this sense, artificial intelligence is a registry of power.” (p.8)

= AI does not exist in a vacuum

- “artificial intelligence is **both embodied and material**, made from natural resources, fuel, human labor, infrastructures, logistics, histories, and classifications.”
- “AI systems are **not autonomous, rational**, or able to discern anything **without extensive**, computationally intensive training with large datasets or predefined rules and rewards.”
- “depends entirely on a **much wider set of political and social structures**. And due to the **capital required** to build AI at scale and the ways of seeing that it optimizes AI systems are **ultimately designed to serve existing dominant interests**. In this sense, artificial intelligence is a **registry of power**.”

Not just a technical thing

Break with the **idea** that AI is **purely technical**

“AI is **technical** and **social**”

AI = just technical domain $\Rightarrow$ **convenient narrative** to hide the underlying issues

- “To understand how AI is fundamentally political, we need to go beyond neural nets and statistical pattern recognition to instead ask *what* is being optimized, and for *whom*, and *who* gets to decide.”

AI is an extractive industry + made of human labor

Earth

- Mineral extraction for computation equipment
- Energy cost for software training

Labor

- invisible work » microtask / click workers
- impacts on work (surveillance)

State and power

- AI used by the state as a tool
- interdependence of tech and military sectors
- outsourcing of state activities to AI companies/systems » outsourced state

Earth

The new oil...



Data Is the New Oil of

BUSINESS

CULTURE

GEAR

IDE

SENETIS

HOME SERVICES WHO WE ARE

If data is the new oil, AI is the refinery & the Cloud is the pipeline

19th February, 2019 | Glenn Vanderlinden



SHARE



DATA IS THE NEW OIL OF THE DIGITAL ECONOMY



... vs Good old mineral extraction

Tom Hegen



AI is based on concrete (mineral) resources

AI/data as new ore etc. » not quite ok but not quite wrong

But it hides the fact that AI needs matter to function.

Possible criticism: it's more the computers than AI itself

- even without AI, we would still extract these minerals to make computers
- but AI does need computers. And is more and more greedy in computing power
- cf. GPU crisis



San Francisco

First golden age

» gold



“The ores extracted from holes in the ground were sold to create the stories in the air; the deeper the **extractions** went, the **higher** the **great towers** of office work stretched into the sky.”

Now

» AI and “gray gold” (lithium crystal)



What resources are being mined / where we mine them

What ?

“short list of twenty-three minerals that are a high “sup- ply risk” to manufacturers”

17 rare earth elements:

- lanthanum, cerium, praseodymium, neodymium, promethium, samarium, europium, gadolinium, terbium, dysprosium, holmium, erbium, thulium, ytterbium, lutetium, scandium, and yttrium. (p.33)

Where ?

Silver Peak, Nevada

Salar south west Bolivia

Central Congo

Baotou (Mongolia, China)

Indonesia

Ecological and political issues

Pollution and **environmental** issues

- Ex. Baotou. Bayan Obo Mines
- 70% of world reserve (china supplies 95% of rare mineral market)
- acid baths, Toxic black mud
- Tin in Indonesia : unregulated work / no environmental protections

Political tension

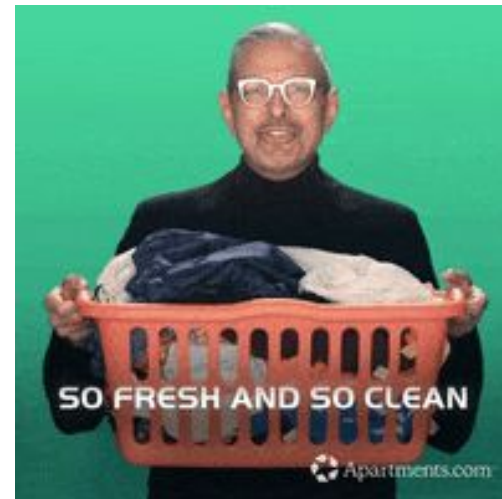
- Tension around environmental destruction
- Colonial heritage / post colonialism » telegraph and submarine cables

Not so clean

Clean tech (vs electrical energy)

- the cloud //// carbon footprint, pollution, etc.
- Putting costs aside and get them out of sight (datacenters, etc.)
- Greenwashing
- etc.

but in fact a huge energy cost



Energy cost

Estimation : Tech sector = **14** % of greenhouse emissions in 2040

NLP model training > 660,000 pounds of carbon dioxide emissions

- 5 cars over their total lifetime / 125 round-trip flights from New York to Beijing. p. 42
- optimistic...

Multiplication of training costs

- Open AI : *10 each year since 2012

Datacenters and ...

- electricity consumption (china 2018 : 73% from coal, 99 millions ton of CO₂)
- **Water** ex NSA, bluffdale, Utah (1.7 million gallons of water/day ?)
- are hidden from view, which minimizes their perceived impact

Megamachine

Origin : “[...] the megamachine to illustrate how all systems, no matter how immense, consist of the work of many individual human actors.”

⇒ “Artificial intelligence is another kind of megamachine, a set of technological approaches that **depend on industrial infrastructures, supply chains, and human labor that stretch around the globe** but are kept opaque.”

Labor

AI and labor

A two-way street

- influence of AI on work (organization, time management, e.g. Amazon)
- but also the **work hidden** behind the “artificial” in AI

Rather than AI replacing humans Crawford looks at

- how humans are increasingly being treated as robots in algo-based work management
- how humans are working for AI

Evolution of the role of AI at/on the job

Before : replacement of **low-skilled** tasks

- e.g.: secretaries

Today: **white collar** replacement ambition

- Automation of judicial decisions, medical decisions, etc.
- ex. Translation

Power asymmetry

- Rhetoric that human AI collab is beneficial
- but this is **not a fair negotiation**
- radiologist has some leeway, not really the case for the amazon employee

Historical perspective : AI / factories

AI implication are not so new:

- Monitoring
- Machine collaboration
- Division and automation of tasks
- Etc.

Examples :

- Historical : Smith, inspection house, panoptic, pointing, etc.
- Current : app tracking movement, data prediction, etc.

Labor behind the AI curtain

Casilli / Tubaro / Gray / Suri / Irani, etc.

- “ghost work.” / “human-fueled automation.” / “travailleurs du clic”
- [The Humans Working Behind the AI Curtain](#), Mary L. Gray and Siddharth Suri

The **labor behind** the “AI”

- poorly paid
- difficult conditions (precariousness / exposure to violence, etc.)
- devalued
- outsourced/delocalized
- invisibilized

Not so artificial

Yet **indispensable** to AI

- “But without this kind of work, AI systems won’t function. The technical AI research community relies on cheap, crowd-sourced labor for many tasks that can’t be done by machines.” p.64
- “crowdsourcing”

Incentives to **ignore** this labor

- you can underpaid them
- it’s better for communication to not show the humans inside. Yep, AI is “magic”

~~“Ceci n’est pas une pipe”~~ vs “this is an AI” (nope)

The **lies** of “AI”

- “40 percent of European startups that are classified as AI companies don’t actually use artificial intelligence in a way that is “material” to their businesses” (a bit more complicated in fact)

Passing off **human work as AI**

AI potempkin

- not really AI, but a bit more than just facades
- potempkin » Illusion/trompe l’œil

Fauxtimation

- overselling systems that in reality are not really automated



Consequences

The **real cost** of AI in human labor is **hidden**

- which allows to devalue it and to pay it less
- “the **true labor costs** of AI are being consistently **downplayed and glossed over**, but the forces driving this performance run deeper than merely marketing trickery. It is part of a **tradition of exploitation and deskilling**, where people must do more tedious and repetitive work to back-fill for automated systems, for a result that may be less effective or reliable than what it replaced.”

It also **relocates** the human work

- “Fauxtimation does **not directly replace** human labor; rather, it **relocates** and **disperses** it in **space** and **time**. In so doing it increases the **disconnection** between **labor and value** and thereby performs an **ideological** function.”

State

Military and intelligence agencies influence

Military **funding** (DARPA, etc.)

- Office naval research and Dartmouth conference

Intelligence agencies and big data pandora's box

- metadata
- Snowden

= technology comes first and the world need to adapt

- “One document notes that the agency was working on multiple fronts to “aggressively pursue legal authorities and a policy framework mapped more fully to the information age.” In other words, change the laws to fit the tools, not the other way around.”
- Similar patterns in AI for health (but with more regulation)

Shaping AI

Military priorities **shaping** AI

- automation, surveillance, computer vision, self driving cars, etc.
- e.g.: interest in the defense sector for some of the developments of start up imaging (image recognition)

“From the point of view of the state, the turn to big data and machine learning expanded the modes of information extraction and informed a social theory of how people can be tracked and understood: you shall know them by their metadata.”

“The military priorities of command and control, automation, and surveillance profoundly shaped what AI was to become. The tools and approaches that came out of DARPA funding have marked the field, including computer vision, automatic translation, and autonomous vehicles.”

Tech/military/public power interdependencies

Interlacing with military research funding

Interconnection between tech and military sectors

- Palantir/google/etc.

“The result is a deep and rapid expansion of surveillance and a blurring between private contractors, law enforcement and the tech sector, fueled by bribes and secret deals.” (p.223)

Project Maven



Tensor flow and detection on images from drones

Not to talk about it. The first rule of ~~fight club~~ Project Maven

Mobilization of employees and withdrawal of google

AI principles



The Outsourced State

From enemies of the state to civilians

- e.g: Palantir. Specific design, role for ICE, etc.
- Conditioned/tracked welfare with AI

France

- using AI to track down Family Allowance Fund (CAF) recipients
- Comment l'intelligence artificielle cible les plus précaires (Tesquet)
- AI for swimming pool declaration fraud (with Capgemini and Google Cloud)

The previous case lends itself to a smile, but what about the case of **social credit**? (we always think of China, but TESQUET shows that it is far from being absent from Western countries, e.g. banking, etc.)

The issue of accountability

Who should be held accountable if AI leads to discrimination, goes off the rails, etc.

What about justice / medical decisions ?

“Currently most states attempt to disclaim any responsibility for problems created by the AI systems they procure, with the argument that “we can- not be responsible for something we don’t understand.” p.199

POWER

Not neutral. Not without human.

“Artificial intelligence is **not an objective, universal, or neutral** computational technique that makes determinations without **human direction**. Its systems are embedded in **social, political, cultural, and economic worlds**, shaped by **humans, institutions, and imperatives** that determine *what* they do and *how* they do it.”

AI = expression of power

“they can **reproduce, optimize, and amplify existing structural inequalities**. This is **no accident**: AI systems are built to see and intervene in the world in ways that **primarily benefit the states, institutions, and corporations that they serve**. In this sense, AI systems are **expressions of power** that emerge from wider **economic and political forces**, created to **increase profits and centralize control** for those who **wield** them. But this is **not how the story of artificial intelligence is typically told**.”

Magic AI / mystification / avoiding responsibility

Algorithmic exceptionalism

- AI as “magic”: masks environmental consequences and human labor, etc.

Black box, too hard to understand, too powerful = removes the need to regulate and the Q° of accountability

- “The fact that deep learning approaches are often uninterpretable, even for the engineers who created them, gives these systems an **aura** of being **too complex to regulate** and **too powerful to deny**.”

Obscuring by mystification

- “employed in public settings to argue for a phenomenon’s inevitability. We are told to **focus on the innovative nature of the method rather than on what is primary: the purpose of the thing itself**. Above all, enchanted determinism **obscures power and closes off informed public discussion, critical scrutiny, or outright rejection**”

Defining AI also means setting a framework

To define things is also to define a way of grasping them and what can (and can not) be contested / criticized / analyzed

- “I’ve argued that there is much at stake in **how we define AI**, what its boundaries are, and who determines them: **it shapes what can be seen and contested.**”
- “The core issue is the deep entanglement of technology, capital, and power, of which AI is the latest manifestation. **Rather than being inscrutable and alien**, these systems are **products of larger social and economic structures with profound material consequences.**”